

RTL to GDSII Physical Design Implementation of an 8-bit ALU

Using OpenLane and SKY130 PDK

Project Type	ASIC VLSI Physical Design
Design	8-bit Arithmetic Logic Unit (ALU)
Front-End	Verilog RTL + Icarus Verilog + EPWave
Back-End	OpenLane + Yosys + SKY130 + Magic + Netgen + KLayout
Final Output	alu_8bit.gds
Student	_____
Register No.	_____
Department / College	_____

Complete Project Report with RTL Code, Testbench, Simulation Waveform and Physical-Design
Screenshots

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1. Abstract

This project presents the complete implementation of an 8-bit Arithmetic Logic Unit (ALU) from Verilog RTL to final GDSII using an open-source ASIC physical design flow. The RTL was functionally verified using Icarus Verilog and EPWave. The verified design was synthesized using Yosys through OpenLane and mapped to SKY130 standard cells. The physical-design flow included floorplanning, power planning, standard-cell placement, routing, and signoff verification. Magic was used for Design Rule Check (DRC), Netgen for Layout Versus Schematic (LVS), and KLayout for layout inspection. The final GDSII was generated successfully. The project evidence included in this report contains the actual RTL/testbench code supplied for the project, the simulation waveform, and the actual floorplan, power-plan, and placement screenshots supplied for documentation.

2. Introduction

An Arithmetic Logic Unit is a fundamental digital datapath block that performs arithmetic and logical operations. In an ASIC flow, an RTL description must be verified and then transformed into technology-mapped standard cells. Those cells are physically arranged, connected to power and signal networks, routed, checked against design rules, compared with the intended netlist, and finally streamed out as GDSII.

The objective of this project is to demonstrate that complete flow for an 8-bit combinational ALU using OpenLane and the SKY130 PDK. The project also demonstrates how the physical layout evolves through floorplanning, power planning, placement and routing.

3. Project Objectives

- Implement an 8-bit ALU with eight arithmetic and logical operations.
- Verify all specified operations using a Verilog testbench.
- Generate and inspect simulation waveforms.
- Synthesize the RTL into SKY130 standard-cell logic using Yosys/OpenLane.
- Perform floorplanning and power-distribution-network generation.
- Perform standard-cell placement and detailed routing.
- Run physical signoff checks including DRC and LVS.
- Generate the final GDSII layout and inspect it using KLayout.

4. ALU Architecture and Specification

The implemented ALU has two 8-bit data inputs, one 3-bit operation-select input, one 8-bit result output, and one carry output. The OP[2:0] code selects one of eight operations.

OP[2:0]	Operation	Function
000	ADD	$A + B$
001	SUB	$A - B$
010	AND	$A \& B$
011	OR	$A B$

OP[2:0]	Operation	Function
100	XOR	$A \oplus B$
101	NOT	$\sim A$
110	INC	$A + 1$
111	DEC	$A - 1$

Top-level interface: A[7:0], B[7:0], OP[2:0] → alu_8bit → RESULT[7:0], CARRY.

5. RTL Design and Source Code

The design uses a combinational always @(*) block and a case statement. RESULT and CARRY are initialized before the selected operation is evaluated. For ADD, SUB, INC and DEC, the concatenation {CARRY, RESULT} is used to retain the extra arithmetic bit in CARRY.

```
`timescale 1ns/1ps module alu_8bit( input [7:0] A, input [7:0] B, input [2:0] OP, output reg [7:0] RESULT,
output reg CARRY ); always @(*) begin RESULT = 8'b00000000; CARRY = 1'b0; case (OP) 3'b000: begin {CARRY,
RESULT} = A + B; end 3'b001: begin {CARRY, RESULT} = A - B; end 3'b010: begin RESULT = A & B; end 3'b011:
begin RESULT = A | B; end 3'b100: begin RESULT = A ^ B; end 3'b101: begin RESULT = ~A; end 3'b110: begin
{CARRY, RESULT} = A + 1'b1; end 3'b111: begin {CARRY, RESULT} = A - 1'b1; end default: begin RESULT =
8'b00000000; CARRY = 1'b0; end endcase end endmodule
```

6. Testbench and Verification Code

The testbench instantiates alu_8bit, applies the eight operation cases, and generates a VCD file for waveform inspection. The supplied testbench uses \$dumpfile and \$dumpvars so the resulting signal activity can be viewed in EPWave.

```
`timescale 1ns/1ps module testbench; reg [7:0] A; reg [7:0] B; reg [2:0] OP; wire [7:0] RESULT; wire CARRY;
alu_8bit uut ( .A(A), .B(B), .OP(OP), .RESULT(RESULT), .CARRY(CARRY) ); initial begin
$dumpfile("alu8bit.vcd"); $dumpvars(0, testbench); // ADD A = 8'd10; B = 8'd5; OP = 3'b000; #10; // SUB A =
8'd10; B = 8'd5; OP = 3'b001; #10; // AND A = 8'b10101010; B = 8'b11001100; OP = 3'b010; #10; // OR OP =
3'b011; #10; // XOR OP = 3'b100; #10; // NOT A = 8'b10101010; OP = 3'b101; #10; // INC A = 8'd255; OP =
3'b110; #10; // DEC A = 8'd10; OP = 3'b111; #10; $finish; end endmodule
```

7. RTL Simulation and Waveform

The RTL was simulated using Icarus Verilog. EPWave was used to inspect the generated waveform. The supplied waveform screenshot shows A, B, OP, RESULT and CARRY over the test sequence. The operation code changes from 0 through 7 and the displayed result values correspond to the expected ALU functions.

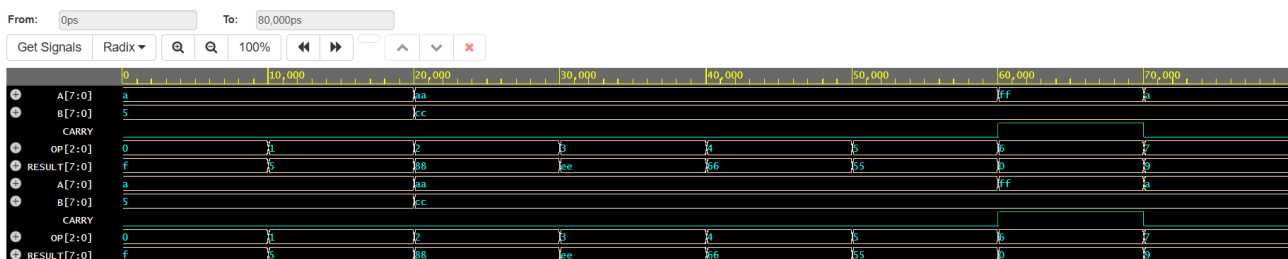


Figure 7.1 — Actual RTL simulation waveform supplied for the project.

Operation	Inputs / Operation	Observed result
ADD	10 + 5	0F (15)
SUB	10 – 5	05 (5)
AND	AA & CC	88

Operation	Inputs / Operation	Observed result
OR	AA CC	EE
XOR	AA ^ CC	66
NOT	~AA	55
INC	255 + 1	00, CARRY = 1
DEC	10 – 1	09

Simulation conclusion: the supplied waveform demonstrates the expected response for all eight operation codes.

8. ASIC Design Flow and Tools

Stage	Tool / Environment
RTL design	Verilog
RTL simulation	Icarus Verilog
Waveform	EPWave
Logic synthesis	Yosys through OpenLane
Physical design	OpenLane
Technology	SKY130 PDK
DRC	Magic
LVS	Netgen
Layout inspection	KLayout
Environment	WSL2 + Ubuntu + Docker

Overall flow: RTL → Simulation → Synthesis → Gate-Level Netlist → Floorplan → Power Planning → Placement → CTS status → Routing → DRC/LVS → Final GDSII → KLayout.

9. Logic Synthesis and Gate-Level Netlist

Yosys synthesis was executed through the OpenLane flow. The RTL was converted into a technology-mapped gate-level netlist using SKY130 standard cells. The synthesized design contains combinational standard-cell logic and no flip-flops, consistent with the clockless ALU architecture.

Metric	Result
Synthesized cell count	174
Total physical cells	572
AND cells	12
NAND cells	8
NOR cells	26
OR cells	72
XOR cells	27
XNOR cells	10

Metric	Result
MUX cells	9
DFF cells	0

Examples of SKY130 cells observed in the gate-level netlist include sky130_fd_sc_hd_and2_2, sky130_fd_sc_hd_or2_2, sky130_fd_sc_hd_nor2_2, sky130_fd_sc_hd_xnor2_2 and sky130_fd_sc_hd_buf_1.

10. Floorplanning

Floorplanning establishes the die/core boundary and the region in which standard cells are arranged. The project used a target core utilization of 40% and an aspect ratio of 1.0. The reported core boundary coordinates were (5.52, 10.88) to (65.32, 70.72).

Parameter	Value
Core Area	3578.432 μm^2
Die Area	0.005844 mm^2
Target Core Utilization	40%
Aspect Ratio	1.0
Core boundary	(5.52, 10.88) to (65.32, 70.72)

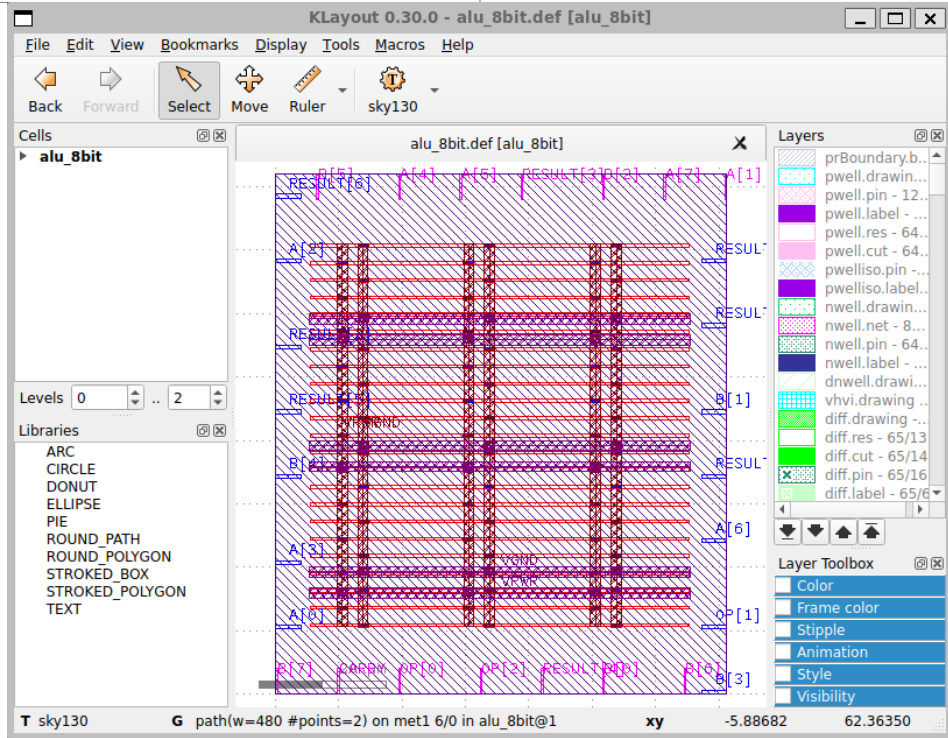


Figure 10.1 — Actual KLayout floorplan screenshot supplied for the project.

11. Power Planning / PDN

Power planning establishes the Power Distribution Network (PDN) for VDD and VSS. The OpenLane flow inserted the standard-cell power grid and reported connectivity for the VPWR and VGND networks. The project flow reported a VDD network voltage of 1.8 V, 349 VPWR PDN nodes and 366 VGND PDN nodes, with all reported PDN stripes connected.

IR-drop note: the run reported that VSRC locations were not explicitly specified and a default source pattern was used. Therefore this report does not present the run as an accurate measured IR-drop analysis.

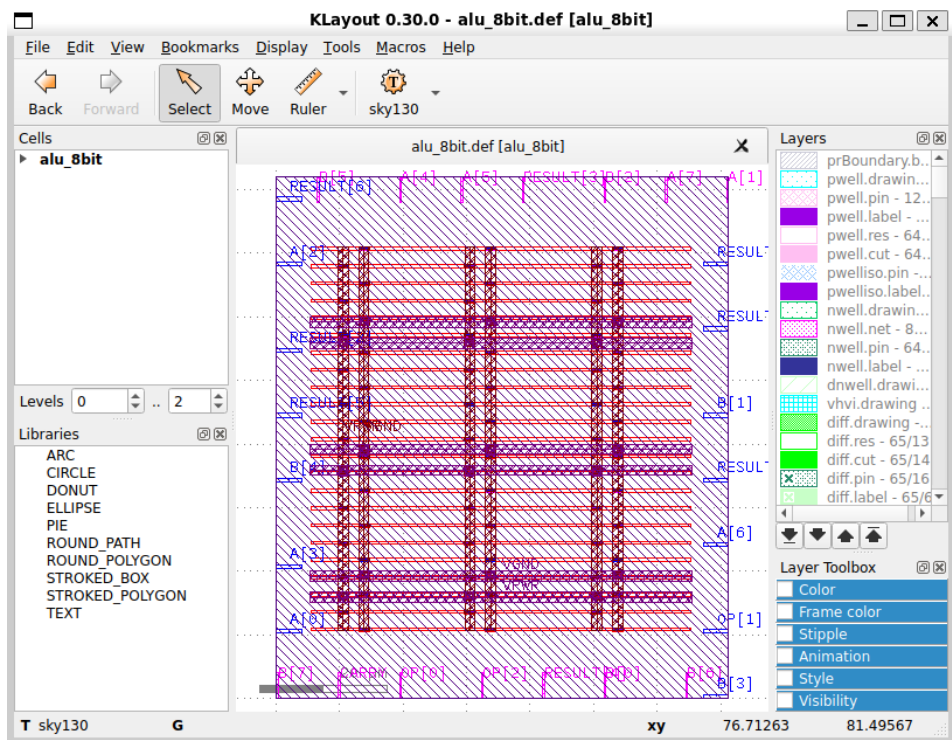


Figure 11.1 — Actual power-plan / PDN KLayout screenshot supplied for the project.

12. Standard Cell Placement

Placement determines the physical locations of synthesized standard cells within the floorplan. Global placement establishes approximate locations and legalization aligns cells to valid standard-cell rows. The physical placement stage also includes supporting cells such as well taps, fill cells and decap cells.

Placement metric	Result
Total cells	572
Welltap cells	48
Fill cells	98
Decap cells	224

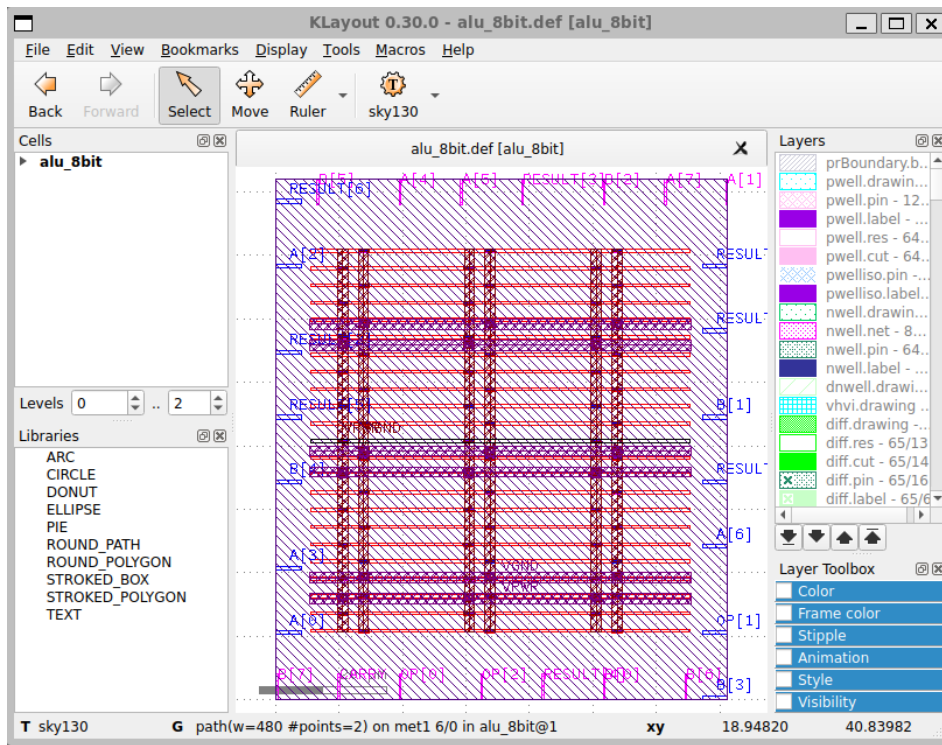


Figure 12.1 — Actual KLayout placement screenshot supplied for the project.

13. Clock Tree Synthesis (CTS)

CTS was not performed for this project. This is intentional and is not a failed stage. The ALU is a purely combinational, clockless design with no clock input and no flip-flops or other sequential elements. Therefore, there is no clock tree to synthesize. The OpenLane configuration used `RUN_CTS 0`.

CTS status: NOT APPLICABLE.

In a clocked sequential ASIC design, CTS would normally construct and balance the clock distribution network. That requirement does not arise for this ALU.

14. Routing

Routing creates the physical metal interconnections between the placed standard cells. The OpenLane detailed-routing stage completed successfully and generated the routing outputs. The reported routing statistics are shown below.

Routing metric	Result
Wire length	4442
Vias	1380
Routing violations	0
M2 utilization	23.25%
M3 utilization	27.50%
M4 utilization	0.39%
M5 utilization	1.11%

The supplied placement/layout evidence can be compared with the routed result during project presentation to explain how physical interconnections are created after placement.

15. Timing Analysis

The flow metrics reported a critical path of 2.64 ns, WNS of 0.0 ns and TNS of 0.0 ns. The metrics also reported a suggested timing period of 10.0 ns. Because the implemented ALU is clockless and combinational, the suggested timing period must not be presented as an actual operating clock, and the project should not claim that the ALU operates at 100 MHz.

16. Design Rule Check (DRC)

DRC verifies whether the physical layout satisfies the geometric and manufacturing rules represented by the technology. The OpenLane signoff DRC report reported COUNT: 0.

DRC RESULT: PASS — 0 violations.

The DRC result is a key physical-signoff indicator because it confirms that the completed layout did not produce reported design-rule violations.

17. Layout Versus Schematic (LVS)

LVS compares the connectivity represented by the physical layout with the intended netlist. The project LVS report states that there were no net, device, pin, or property mismatches and reports Total errors = 0.

LVS RESULT: PASS — 0 errors.

18. Final GDSII and KLayout

After physical implementation and verification, the final GDSII stream-out was generated. The final file is named alu_8bit.gds and has a reported size of approximately 819 KB. The final GDSII was opened in KLayout for physical inspection.

Final GDSII: alu_8bit.gds

Reported file size: approximately 819 KB

Final output location: results/final/gds/alu_8bit.gds

19. Project Metrics

Metric	Value
Core Area	3578.432 μm^2
Die Area	0.005844193625 mm ²
Synthesized cells	174
Total cells	572
Wire count	293
Wire bits	316
Wire length	4442
Vias	1380
Routing violations	0
Magic violations	0

Metric	Value
LVS total errors	0
Pin antenna violations	0
Net antenna violations	0
Critical path	2.64 ns
WNS	0.0 ns
TNS	0.0 ns
Suggested timing period	10.0 ns
Final GDSII	Generated (~819 KB)
Runtime	35 s
Peak memory	532 MB

20. Results and Discussion

The complete project flow successfully transformed the verified 8-bit ALU RTL into a physical ASIC layout. The actual supplied evidence demonstrates the RTL waveform and physical floorplan, power-plan, and placement stages. The OpenLane run produced routing outputs and final signoff results.

The most important signoff results are zero reported routing violations, zero Magic violations, zero DRC violations, zero LVS errors, zero pin-antenna violations, and zero net-antenna violations. The final GDSII was generated successfully.

The project also illustrates why the applicability of a physical-design stage depends on the circuit architecture. CTS is appropriate for clocked sequential logic, but the present ALU has no clock or sequential elements, so CTS is intentionally skipped.

21. Limitations and Notes

- CTS is not applicable to the clockless combinational ALU.
- Accurate IR-drop measurements are not claimed because explicit VSRC locations were not specified.
- The suggested 10 ns timing period is not an actual ALU operating clock.
- The report includes the actual screenshots supplied in this conversation for waveform, floorplan, power plan and placement.
- Additional evidence such as the final routed KLayout view, DRC report and LVS report can be added to the final printed/submitted copy if available.

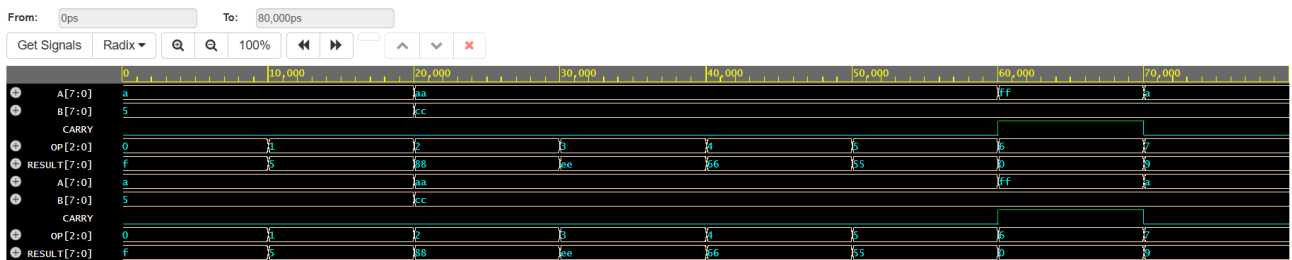
22. Conclusion

The 8-bit ALU was successfully implemented from Verilog RTL to GDSII using OpenLane and the SKY130 PDK. The supplied RTL and testbench verify the eight required operations, and the simulation waveform demonstrates the expected operation sequence. The back-end flow completed synthesis, floorplanning, power planning, placement and routing, followed by physical signoff checks. DRC reported zero violations and LVS reported zero errors. The final alu_8bit.gds file was generated and inspected using KLayout.

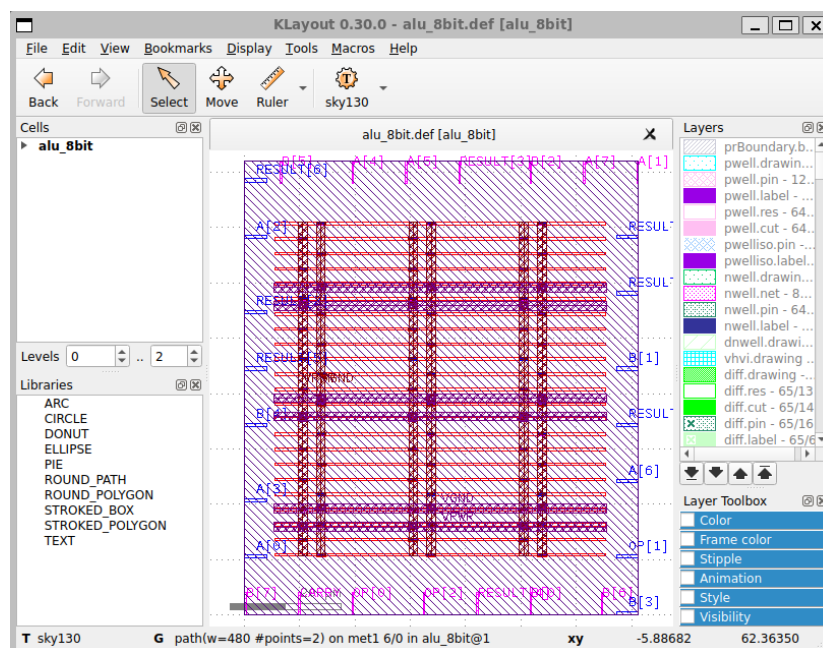
23. Future Scope

- Add additional ALU status flags such as zero, sign and overflow.
- Build a clocked wrapper or larger datapath so CTS and sequential timing closure can be demonstrated.
- Experiment with different utilization, aspect-ratio and placement configurations.
- Define explicit voltage-source locations and perform more rigorous IR-drop analysis.
- Integrate the ALU into a larger datapath or processor and repeat the RTL-to-GDSII flow.

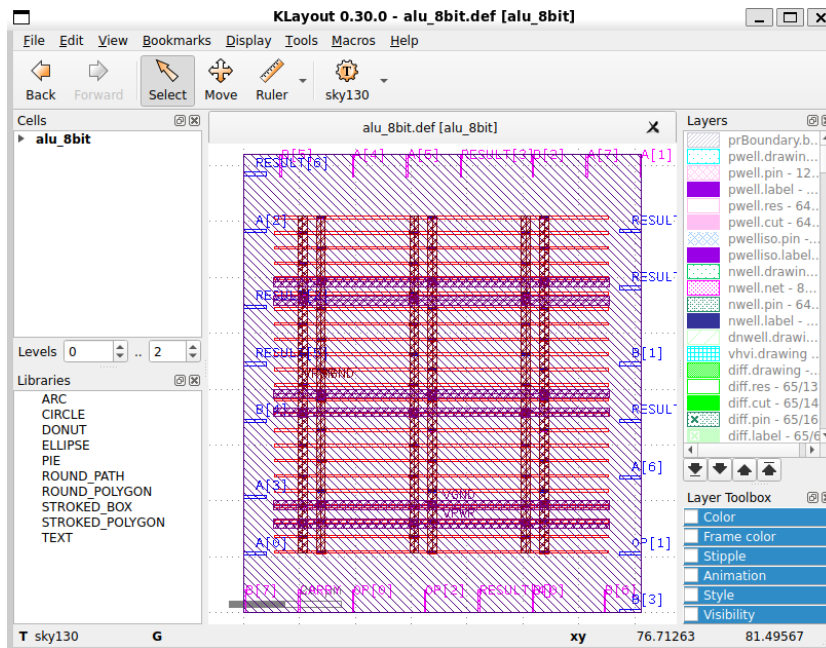
Appendix A — Evidence Screenshots Included



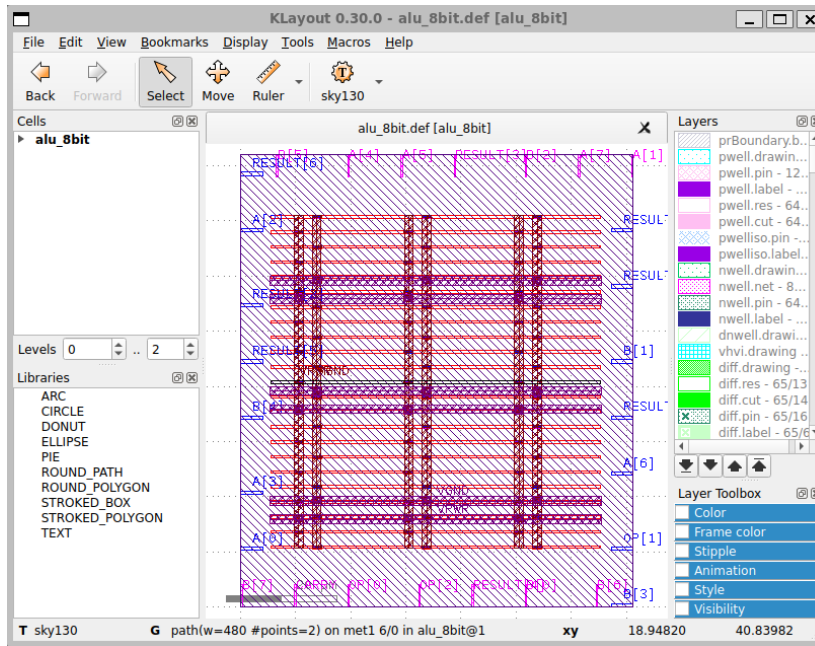
Appendix Figure A.1 — RTL waveform



Appendix Figure A.2 — Floorplan



Appendix Figure A.3 — Power Plan / PDN



Appendix Figure A.4 — Placement

Appendix B — Key Output Files

results/synthesis/alu_8bit.v — synthesized gate-level netlist

results/placement/alu_8bit.nl.v — placement netlist

results/routing/alu_8bit.def — routed DEF

results/routing/alu_8bit.odb — routed OpenROAD database

reports/signoff/drc.rpt — signoff DRC report

reports/signoff/37-alu_8bit.lvs.rpt — LVS report

results/final/gds/alu_8bit.gds — final GDSII